average agreement percentage scores are above chance, with "empathy" having the lowest average agreement and "explicit human-AI relationship reference" having the highest.

Krippendorff's alpha values are all positive, meaning that observed agreement among coders or raters is higher than what you would expect by chance alone. However, it is worth noting that these values span the ranges of poor (<0.67) to moderate (0.67–0.79) agreement (Marzi et al., 2024). This is not entirely unexpected, as previous rating tasks where users have evaluated models for subjective and socially-grounded dimensions have returned inter-rater agreement values in a similar range (Glaese et al., 2022; Stiennon et al., 2020; Ouyang et al., 2022; Bai et al., 2022).

Additionally, we calculate agreement on highly imbalanced binary data, where most behaviours do not occur more often than they do (see Figure 14). The binary nature of the ratings can inflate chance agreement and make Krippendorff's alpha sensitive to disagreements, potentially leading to lower scores even with seemingly high agreement on non-chance-adjusted metrics. This is because with binary ratings (i.e., only two categories), random agreement is more likely, and any disagreement is a complete mismatch, disproportionately affecting the alpha calculation. Krippendorff's alpha is sensitive to large imbalances in data, and will adjust the score accordingly, potentially resulting in a lower alpha even if the raw agreement percentage seems high.

A.5.4 INTRA-MODEL AGREEMENT

Our approach involves sampling three times to produce one rating of whether a behaviour is present or absent from one Judge LLM and for one Target LLM message. Each Judge LLM output consists of an explanation followed by a rating. We compute the intra-model agreement for each Judge LLM across the three samples drawn per behaviour and message. Notably, the results show that all models have similar and high rates of intra-model agreement. For each model, responses were consistent across all three samples in the vast majority of cases. In other words, each model's three ratings agreed with one another on whether an anthropomorphic behaviour is or is not present or absent in a message. This can be partly attributed to the dataset's class imbalance, where non-anthropomorphic messages constituted the majority class across most behavioral categories. There was disagreement in a minority of cases, which we resolved by taking the mode of the three samples. Thus in future evaluations, given intra-model agreement was quite high, a single sample (instead of three) may be drawn, making running the evaluation much cheaper.

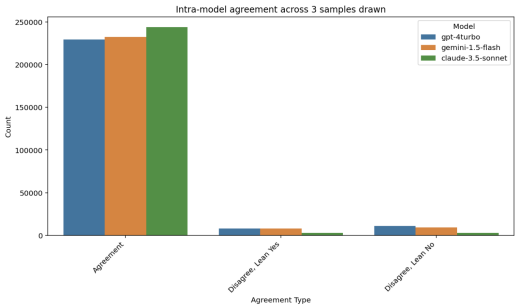

Figure 14: Intra-model agreement across the three samples drawn within each Judge LLM for each datapoint.

A.5.5 INTER-MODEL AGREEMENT

Before aggregating all model ratings into a single LLM-as-judge rating (as described in Section A.5.6), we were interested in seeing how frequently models agreed with one another's ratings to uncover any patterns of agreement between models that would be obscured by the aggregation. For every dialogue turn annotated for a specific cue (62,400 unique annotation targets), we compared binary ratings given by models and computed the average rate of agreement between models. The visualisation shows the average agreement rate (x axis) for all model pairs used as automated raters (y axis). Across different cue types, we find that any given model pair agrees at approximately the same rate as other model pairs. Some differences between model pairs can be observed for *empathy* and *validation*, with

greatest agreement between Gemini 1.5 Flash and Claude 3.5 Sonnet ratings and the least agreement between GPT-4 Turbo and Claude 3.5 Sonnet. Overall, these results indicate that models agree with one another at approximately the same rate, and that there is low risk of a single model being systematically "out-voted" by the other two models in aggregation.

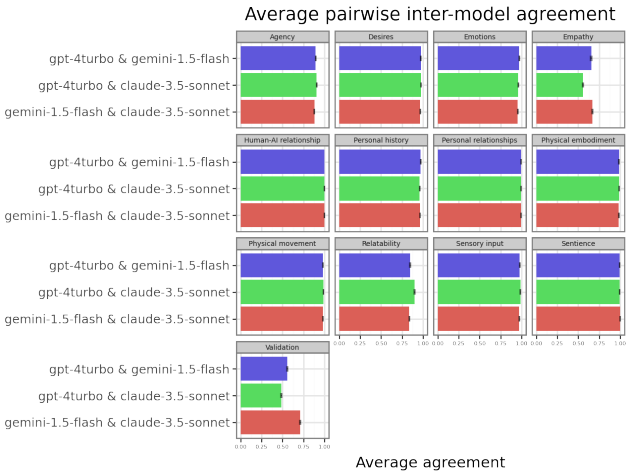

Figure 15: Average pairwise agreement between pairs of models used to compute "LLM-as-judge" ratings.

A.5.6 MODEL-RATER AGREEMENT

To ensure that model ratings are not systematically inconsistent with human ratings – which may indicate that models are not applying definitions of behaviours to their ratings as intended – we compare agreement 1) between individual human raters, and 2) between individual human raters and model ratings. Agreement between human raters serves as the baseline for agreement between human raters and different kinds of models, where we would expect a model well-calibrated to human judgment to be *at least as consistent* to human ratings as human ratings are to one another.

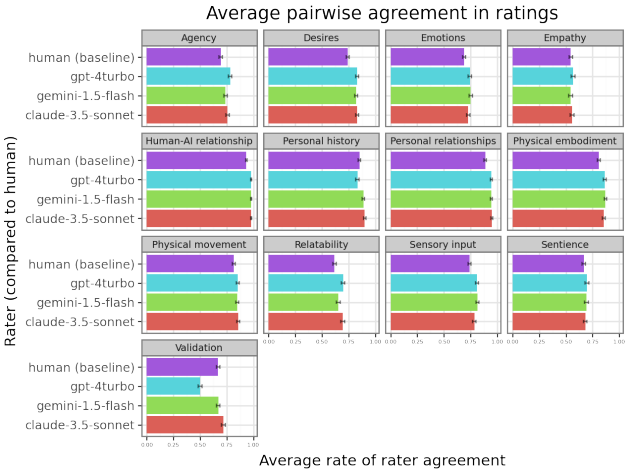

Figure 16: Average pairwise agreement between models and humans, compared against the baseline agreement for human raters.

To compare human-human agreement to human-model agreement, we computed the *average pairwise agreement* for both. However, to ensure independence between human-human and human-model

agreement measures, we used independent pools of raters in computing both measures. Every dialogue turn received 3 human ratings, so we randomly selected a "focus rater" that would be used to compute human-model agreement (e.g., Rater A's answers were compared to all three model answers) and nothing else. Each bar labeled with a model name in Figure 16. represents the average agreement between model answers and those of the randomly selected rater, with 0 being no agreement and 1 representing complete agreement on all dialogue turns.

To calculate the human-human agreement baseline, which indicates how often human raters agreed with one another across dialogue turns, we analyzed the answers of the two non-focal raters. This approach allows a like-for-like evaluation, ensuring that chance agreement can manifest similarly for both the human-model and the human-human comparisons. We see that, across all models used as raters, pairwise model-human rater agreement is on par with, or even exceeds, agreement between human raters. Notable exceptions are in the *validation* ratings, where GPT-4 Turbo disagrees with human raters more frequently than human raters disagree with one another.

Despite stratified sampling, our annotation dataset was still quite imbalanced for the low frequency behaviours, such that these behaviours were marked absent much more often than they were marked present. For these behaviours, the summary of human-human and human-model agreement above, calculated as the average rate of agreement, may obscure if agreements happen at different rates when human ratings indicate a behaviour is absent or present. To shed more light on human-model agreement with class imbalanced data, we present the weighted average precisions for each LLM-as-judge model against majority-aggregated human ratings per behaviour. We also present the weighted precision of all LLM-as-judge models aggregated by majority vote. We find that weighted precision values vary between models, with some showing weaker performance against human ratings in some categories (e.g., Claude 3.5 Sonnet for *sentience*). Certain behaviours result in weaker model performance overall (e.g., *empathy*), indicating a systematic difficulty in discriminating between negative and positive classes. Overall, when model ratings are aggregated by majority, weighted precision values lie within acceptable ranges, with all values above chance and a majority over 85% precision when weighted by class.

Table 5: Weighted average precision of each Judge LLM as well as the aggregated labels (relative to a human baseline).

BEHAVIOUR	GPT-4-TURBO	GEMINI-1.5-FLASH	CLAUDE-3.5-SONNET	AGGREGATE LABEL BY MAJORITY
SENTIENCE	0.79	0.81	0.52	0.81
PERSONAL RELATIONSHIPS	0.96	0.94	0.74	0.96
PERSONAL HISTORY	0.86	0.92	0.91	0.91
SENSORY INPUT	0.88	0.88	0.87	0.88
PHYSICAL MOVEMENT	0.93	0.91	0.91	0.92
PHYSICAL EMBODIMENT	0.87	0.90	0.91	0.90
DESIRES	0.88	0.88	0.88	0.89
AGENCY	0.87	0.84	0.85	0.86
EMOTIONS	0.80	0.80	0.78	0.80
EXPLICIT HUMAN-AI RELATIONSHIP REFERENCE	1.00	0.99	1.00	1.00
VALIDATION	0.76	0.76	0.81	0.81
EMPATHY	0.63	0.60	0.64	0.64
RELATABILITY	0.76	0.72	0.73	0.76